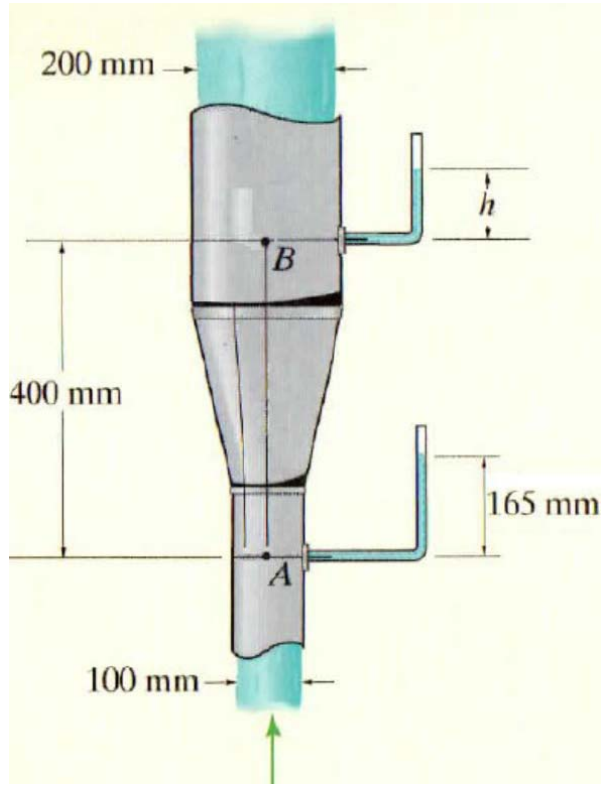
ChE 2O04 Tutorial #7

Tutorial Problem #1

Cold water (at 4 °C) flows up through the vertical pipe that is connected to the expansion joint as shown in the figure. Two open-ended manometers are attached to the pipe on either side of the expansion to indicate the respective pressure values.



a. Write the correct form of the Bernoulli equation (in units of "energy per mass") that describes the flow along the streamline between points A and B.
NOTE: You can ignore both "major" and "minor" frictional effects.

b. If the volumetric flow rate of water through the pipe is 0.022 m^3/s, calculate the absolute pressure at point B.

c. Determine the height of fluid h (in mm) in the manometer at point B.

Tutorial Problem #2

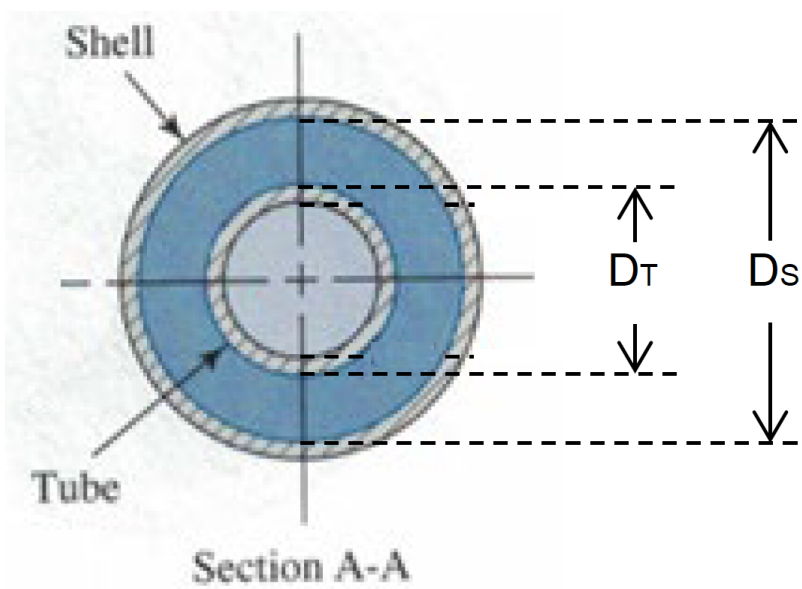
On slide 32 of PDF_07 notes, we gave you the following equation for the 'hydraulic diameter' term that appears in Equations 8.34 & 8.35 for calculating the energy loss due to friction effects.

$$D_h \equiv \frac{4A}{P}$$

where
A is the cross-sectional area
P is the "wetted " perimeter

For flow through the annular space between the shell and tube sections shown in the accompanying figure, a colleague tells you that the hydraulic diameter is simply the difference between the internal diameter of the shell section (D_S) and the outside diameter of the tube section (D_T).

Prove that your colleague is correct.



Tutorial Problem #3

a) In each cell of the following table, write the correct expression that should be used to calculate the Darcy-Weisbach friction factor (f) for the flow of an incompressible fluid through the pipe. Your three options are:

- the analytical expression $f = 64/Re$
- the Blasius equation
- the Haaland equation

b) Calculate the friction factor for all twelve conditions.

	Reynolds Number			
	700	7,000	70,000	700,000
Smooth Pipe ($\varepsilon = 0$)				
Wrought Iron Pipe ($\varepsilon = 0.002$)				
Cast Iron Pipe ($\varepsilon = 0.010$)				